Edward A. Varvak

Department of Mathematics
University of Chicago
5734 S University Ave
Chicago, IL 60637

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Dear Members of the Hiring Committee,

I am a graduate student at the University of Chicago, working in complex algebraic geometry and Hodge theory under the direction of Benjamin Bakker, and I expect to receive my Ph.D. in May 2027. I am applying for the position of Benjamin Peirce Fellow in Mathematics (Harvard-BPFELLOW20272028). My research concerns the monodromy and Hodge theory of Lagrangian fibrations on hyper-Kähler and other symplectic varieties, as well as moduli spaces of abelian varieties and their Mumford–Tate groups. At Harvard, I would be particularly interested in working with Mihnea Popa and Mark Kisin.

My main goal in teaching is for students to leave a course able to explain why the methods they learned work, not only how to carry them out. As such, I try to show them what a formula is doing before they use it. For example, when teaching calculus at the University of Chicago, I go through volumes of revolution by drawing the solid, sketching a typical slice, and working out its area with the class, instead of handing out the washer and shell formulas.

When I work through a proof or an example, I make a point of aking the class what the next step should be and waiting for a guess before going on. I find that doing so creates an environment for the class where the students are unafraid to ask questions of their own accord. I also try to match the material I cover to the level of the class. At the University of Illinois Chicago, most of my students in Finite Mathematics for Business took it to satisfy a requirement, and many arrived unsure of their mathematical ability; with them, I set aside more advanced applications in favor of making sure every student could follow.

Outside the classroom, my advising so far has been informal. In the learning seminars I organized, I helped speakers prepare their talks. While I would be happy to continue organizing seminars at Harvard, I would also like to try out a more concrete mentoring arrangement, such as advising undergraduate reading projects in algebraic geometry. Likewise, I would welcome the chance to teach a one-term course in algebraic geometry or Hodge theory.

Along with this letter, I include my curriculum vitae, research statement, and service statement. Research letters will come from Benjamin Bakker (Chicago), Radu Laza (Stony Brook), İzzet Coşkun (UIC), and Valentino Tosatti (Courant), and a teaching letter from McKinley Meyer (UIC). Thank you for your consideration.

Sincerely,

Edward A. Varvak